

**Problem Statement ID – SIH26055**

**Problem Statement Title –  
Smart Scan Strategy for Electronic Warfare**

**Theme – Robotics and Drones**

**PS Category – Software**

**Team ID – 157808**

**Team Name – Axiom X**



# Intercept the right band at the right time.

Spectra Scheduler learns from receiver hits and misses to decide which frequency band to observe, when to revisit it, and how long to listen—without reliable prior emitter intelligence.

## CHALLENGE

### Fixed sweeps miss windows

Wide spectrum; narrow receiver.  
Brief activity between revisits.

## SOLUTION

### Learn from hits and misses

Forecast activity from scan history.  
Adapt band choice and dwell.

## DIFFERENTIATOR

### Coordinate frequency and time

Timing forecasts guide listening.  
Retune cost and coverage matter.

## Observe

Listening mask + detections

## Predict

Activity and intercept time

## Schedule

Band + listening dwell

## Listen

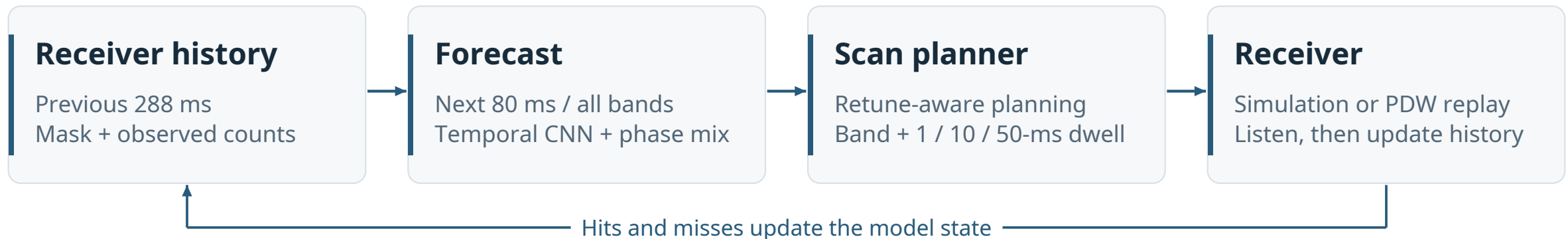
Retune, then hold

Learn from each revisit

## Predict activity. Choose the next scan.

Selected scheduler: causal temporal forecasts with retune-aware planning.

Repository: [github.com/Siwach777/spectra-scheduler](https://github.com/Siwach777/spectra-scheduler)



**OUTPUTS** Capture probability • intercept time • interception ratio

**CAUSAL INPUTS** No emitter identity • no supplied period / phase • no future information

### Simulation + learning

Python / NumPy  
PyTorch / CUDA

### Pulse replay + data

HDF5 / h5py  
scikit-learn

### API + visualization

JSON / HTTP  
JavaScript / Canvas

# Validated in simulation and external pulse replay.

Matched receiver budgets, stronger controls and a working closed-loop demo.

## ENGINEERING

### Receiver simulation

Agile, spatial and periodic emitters.  
Noise, sensitivity and retuning.

## LEARNING

### Trained scheduler

Causal forecasts and scan control.  
CUDA training; held-out checks.

## DATA AND SOFTWARE

### Replay + analyst console

32 unseen recordings evaluated.  
Replay CLI + trained-policy GUI.

## EXTERNAL SYNTHETIC TSRD VALIDATION • 32 RECORDINGS

# 4.16× mean capture versus fixed sweep

Timing 43.49% • fixed sweep 10.45% • emitter discovery 93.35%

## 300-world simulation

Agile / spatial / periodic capture:  
34.37% / 51.56% / 67.66%

## Capture / discovery

Rate-probe: 51.66% capture;  
84.39% emitter discovery.

## Validation boundary

External synthetic data.  
No RF hardware validation.

# Adaptive surveillance. Measurable interception.

Implemented scheduling, prediction and evaluation for Electronic Support receivers.

## SURVEILLANCE

### Capture brief opportunities

Forecast activity before revisits.  
Adapt to agile and periodic emitters.

## RECEIVER RESOURCES

### Use receiver time well

Choose band and dwell together.  
Account for retuning and coverage.

## DEVELOPMENT

### Evaluate faster

Reuse RF worlds and pulse replay.  
Compare policies on matched data.

## CORE CAPABILITIES IMPLEMENTED

### Learned scan scheduling

Band + dwell control from causal receiver history.  
Intercept-time and interception-ratio predictions.

### Simulation, replay and evaluation

Agile, spatial and periodic emitter scenarios.  
Required figures of merit + physical-time replay.

Warmed serial scheduler p99: 0.464–0.532 ms on CUDA; receiver I/O and cold startup excluded.

# Research, source code and demonstration.

Primary research sources and project links.

- |           |                                                  |                                                                                                               |
|-----------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>01</b> | <b>Radar data</b><br>arXiv:2602.03856            | <b>Gunn et al. — The Turing Synthetic Radar Dataset</b><br>Pulse descriptor words for association and replay. |
| <b>02</b> | <b>Dataset contract</b><br>Official dataset card | <b>The Alan Turing Institute — TSRD dataset card</b><br>Receiver modes, feature units and data splits.        |
| <b>03</b> | <b>Temporal prediction</b><br>arXiv:1803.01271   | <b>Bai, Kolter &amp; Koltun — Sequence Modeling</b><br>Causal temporal convolution for activity forecasting.  |
| <b>04</b> | <b>Policy learning</b><br>arXiv:2402.14740       | <b>Ahmadian et al. — Back to Basics / RLOO</b><br>Trajectory-level learning for scan decisions.               |

## SOURCE CODE • GITHUB

[github.com/Siwach777/spectra-scheduler](https://github.com/Siwach777/spectra-scheduler)

## DEMO VIDEO • YOUTUBE

[youtu.be/G0SuhfmfhfM](https://youtu.be/G0SuhfmfhfM)

METRICS Detection • false alarms • sensitivity • intercept rate • reward / cost • accuracy • timing error